Smart Student Presence Monitoring System

Problem Statement:

Manual attendance tracking and academic record management is **time-consuming** and prone to errors.

Teachers find it difficult to maintain and analyze records, while students lack easy access to their attendance and performance data.

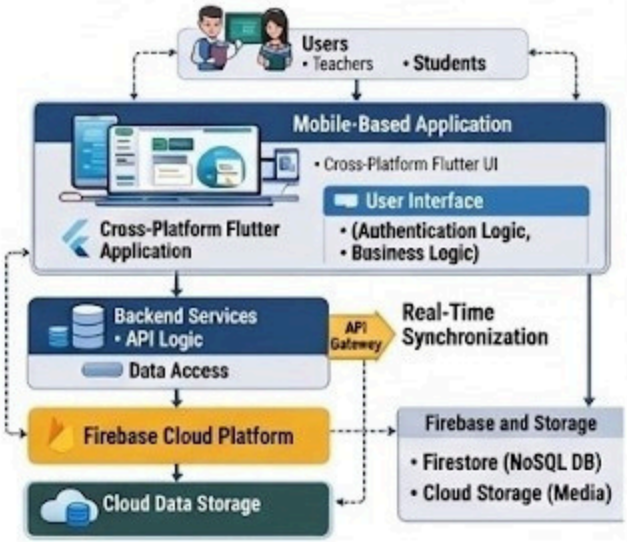
Objectives:

- To manage and store student data in a centralized system
- To digitize attendance tracking and reduce manual work
- To monitor attendance and send alerts below 75%
- To generate reports and analyze academic performance

Work Flow:

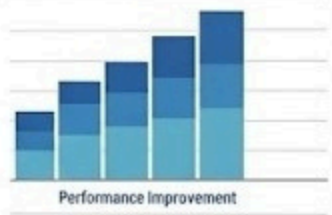
1. User logs in (Teacher/Student)
2. Teacher marks attendance & update marks
3. Data stored in Firebase database
4. System processes data
5. Reports & analytics generated
6. Students view records and performance

Architecture / Block Diagram



Results:

Results & Analysis



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- Improved Data Accuracy
- Real-Time Access for Students
- Easy Report Generation
- Better Performance Analytics

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Key References:

- Flutter Documentation - https://flutter.dev
- Firebase Documentation - https://firebase.google.com/docs
- Student Information System research papers

Conclusion:

The system provides an efficient and accurate way to manage attendance and academic records digitally.

It improves productivity for teachers and offers easy access and transparency for students.